

Abdoulay Lashley

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EDUCATION

University of Maryland, Robert H. Smith School of Business

College Park, Maryland

Master of Science in Information Systems, GPA: 3.7

August 2024 – December 2025

- **Relevant Coursework:** Data Mining & Predictive Analytics, Designing Generative AI Systems, AI for Business Applications, Data Processing, Analysis, Modeling & Visualization, Database Management, Cloud Computing, Project Management, Industry Practicum
- **Leland – Full-Stack Product Management Bootcamp** (Certificate), 2025
- **AWS Academy Graduate** – Cloud Architecting & Machine Learning Foundations (Certificate), 2025
- **Terrapin Scholar** – University of Maryland, 2024–2025 | **Dean’s List** – 2020–2021, 2022–2023, 2024–2025

University of Maryland, A. James Clark School of Engineering

College Park, Maryland

Bachelor of Science in Bioengineering, GPA: 3.2

August 2018 – May 2023

TECHNICAL SKILLS

Programming & Data Processing:

Python, SQL, R, JavaScript, TypeScript, PySpark MLlib, Spark Streaming, Node.js, Express.js, ETL/ELT, Data Pipelines

Machine Learning & Statistical Analysis:

PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLP, LLMs, OpenAI API, Computer Vision (CNNs), Transfer Learning, Grad-CAM, Model Fine-tuning, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Regression & Classification, Clustering, A/B Testing, MLOps, AWS Comprehend

Business Intelligence & Data Visualization:

Power BI, Tableau, Amazon QuickSight, Plotly, Seaborn, ggplot2, Google Analytics, React.js, Next.js, Tailwind CSS, Leaflet.js

Cloud Technologies & Big Data:

AWS (EC2, S3, RDS, Lambda, VPC, CloudFront, Glue, CloudFormation, IAM), Amazon SageMaker, BigQuery, Snowflake, Apache Spark, Apache Hadoop, Microsoft Azure, Google Cloud Platform (GCP)

Databases:

NoSQL, MySQL, MongoDB, Neo4J

Project & Product Management:

Agile, SCRUM, Product Management, Risk Management, QA, JIRA, Confluence, Microsoft Project, Figma (UX/UI) Design

Business Strategy & Communication:

Business Strategy, Stakeholder Communication, Cross-functional Collaboration, Prioritization

Software Tools & Platforms:

FastAPI, Git/GitHub, Jupyter, VSCode, Google Colab, HTML, CSS, Spark GraphFrames, Kafka, Sqoop, OpenCageData API, Geolib, REST APIs

WORK EXPERIENCE

The Brite Group

Reston, Virginia

Data Science Intern – Computer Vision, PyTorch, XGBoost, FastAPI, Next.js, TypeScript, GCP (GPU)

March 2026 – June 2026

- **Architected PulmoVision AI**, a production-ready explainable AI platform for multi-label chest X-ray classification across **14 thoracic conditions**, combining DenseNet-121, ResNet-50, and Inception-V3 CNN backbones with SVM-GAP classifiers in a stacked generalization framework trained on **223K+ chest radiographs** from the CheXpert dataset
- **Engineered a 5-fold, view-aware ML pipeline** generating out-of-fold predictions across **10 model streams** (frontal + lateral) to construct a **141-feature input vector** for an **XGBoost meta-learner**, achieving a **mean OOF AUROC of 0.787** across all 14 labels with a **macro recall of 0.733**, prioritizing clinical sensitivity across pathological conditions
- Identified DenseNet-121 VisualCheXbert (frontal) as the highest-performing **base model at 0.879 mean AUROC**, with **per-label AUROC reaching 0.942**, and implemented a dual-model CAM system fusing CheXbert and VisualCheXbert feature maps with weighted averaging — resolving CUDA instability from backward-pass gradient computation while preserving full spatial explainability
- **Delivered a full-stack clinical diagnostic interface** using **FastAPI and Next.js** with real-time CAM visualization, AUROC-weighted abnormality scoring, plus a clinical assistant powered by **GPT-5.5, Claude 4.7 & 4.6, Gemini 3.1, and Llama 4 Maverick** for diagnostic support featuring 4-tier urgency triage (Critical / Urgent / Routine / Normal) with condition-specific intervention guidance

L’Oréal USA

Somerset, New Jersey

Industrial Quality CO-OP – Product Launch, Data Visualization & Analysis, Process Improvement

June 2023 – August 2024

- **Designed and rolled out eCLAIR digital product**, building consensus across QA, operators, and engineering teams, **reducing micro-contamination incidents by 20%**.
- **Digitized 25 production forms, streamlining accessibility** for line operators and **reducing documentation errors by 30%**, enhancing operational efficiency and regulatory compliance.
- **Built Power BI dashboards that boosted efficiency by 15% and cut downtime by 10%**, presenting insights to **senior leadership** to align stakeholders on data-driven decisions.

PROJECT EXPERIENCE

Veteran Sentiment Analysis Platform – Deloitte GPS | Python, NLP, FastAPI, LLMs

January 2025 - December 2025

- Led a **Deloitte-sponsored** MSIS Capstone project as **Project Manager and Business Requirements Lead**, guiding a **5-person** team to analyze **over 1,000** public Reddit posts from U.S. veterans on key issues such as mental health, financial stress, and educational benefits.
- **Designed and implemented** an **end-to-end NLP pipeline** in Python to clean and normalize text, score sentiment using both traditional models and LLM-based methods, extract key phrases, and cluster conversations into themes and inferred veteran personas.
- **Delivered** an **interactive web dashboard and AI Copilot** built with **HTML, CSS, JavaScript, Chart.js, and a FastAPI-based RAG** service, enabling nontechnical stakeholders to explore sentiment trends, drill into supporting posts, and ask context-aware questions grounded in the data.

Generative AI Dynamic Travel Itinerary Planner – *Python, OpenAI API, FastAPI, JavaScript*

October 2025

- **Designed and developed** a **Generative AI travel planner** that generated **personalized day-by-day** itineraries using **real-time flight, event, and weather data** combined with user preferences such as budget, interests, and trip length.
- **Integrated** the **OpenAI API with Python (FastAPI) backend** and an interactive **HTML/CSS/JavaScript frontend**, enabling natural-language trip modifications like “add more food options” or “reduce my budget.”
- **Contributed to LLM prompt design and orchestration**, ensuring contextual accuracy, ethical use, and transparent explanations, demonstrating a **scalable, user-centric AI application** that improved trip personalization efficiency by **~30% in testing**.

NBA Team Performance Evolution – *Tableau, Python, nba_api, Data Visualization*

October 2025

- **Built an interactive Tableau** story analyzing a decade of NBA team performance (2014–2024) using **data engineered in Python** from the official **NBA Stats API**.
- **Developed** custom metrics for **Offense Index, Defense Index, and Top Teams Metric**, ranking **30 franchises** by efficiency balance to **uncover patterns** behind championship success.
- **Designed 3 dynamic dashboards** and 1 supplemental analytics page, integrating **calculated fields, LOD expressions, and geographic mapping** to visualize the **NBA's transition** into the pace-and-space era.

LAPD Crime Prediction & Analysis – *PySpark, Random Forest, MLlib*

May 2025

- **Built** a scalable PySpark pipeline to analyze **2M+ crime records** (2023–2025) for type **classification, hotspot detection, and spatiotemporal pattern clustering**.
- **Achieved 88.79% top-5 accuracy** using a **Random Forest model** tuned with **CrossValidator** and **grid search**, integrating **StringIndexer, OneHotEncoder, and VectorAssembler** for preprocessing.

Capital Area Food Bank AI Assistant – *React.js, Node.js, OpenAI API, OpenCageData API, JavaScript, JSON*

April 2025

- **Led end-to-end development** of a multilingual AI chatbot for food-insecure neighbors in the **DC metro area**, enabling users to locate and filter **500+ food resources** by ZIP code or location, while addressing barriers like confusing maps, language gaps, transportation access, service requirements, and social stigma using **GPT-3.5** and **geospatial intelligence**.
- **Built** a responsive dark-mode front-end with React.js featuring map-based food bank pins and interactive site cards displaying address, open hours, and service information.
- **Developed** backend RESTful APIs with Node.js/Express.js and OpenCageData API, achieving **>90% match accuracy** and **reducing query latency by 40%** by optimizing JSON schema design and geolocation handling.
- **Cleaned and unified 6+ CAFB datasets** using JavaScript to merge appointment requirements, food format, cultural services, and transportation access metadata into a streamlined structure.
- **Enhanced accessibility** with a custom TranslationService supporting **dynamic English-Spanish language switching** and fallback to GPT for non-location-based queries, significantly improving engagement among **Spanish-speaking users**.

The Weeknd Effect Streaming Analytics – *Python, Pandas, Scikit-Learn, SHAP*

December 2024

- **Engineered multi-platform datasets** and analyzed **200+ audio and release features** to model drivers of global streaming performance.
- **Built regression models** achieving an **R² of 0.82** and applied **SHAP interpretability** to surface key contributors to virality and long-tail engagement.
- **Created narrative data visuals** that clarified **model insights for non-technical audiences**, highlighting patterns such as 3.7B streams for “Blinding Lights.”

Buccal Pulse Oximeter to Overcome Racial Biases – *Python, MATLAB, Arduino*

May 2023

- **Led** a team of five to develop a **buccal pulse oximeter**, reducing **racial biases** in oxygen saturation measurement by **20%** using advanced **sensor technology** and **signal processing**.
- **Developed and optimized** data analysis in **Python and MATLAB**, achieving **95% measurement accuracy** while maintaining **production costs under \$40 per unit**.
- **Demonstrated** the prototype to **20+ clinicians and engineers**, validating its **effectiveness in clinical trials**.

AFFILIATIONS & ADDITIONAL INFORMATION

- **Member** – Black Engineers Society, National Society of Black Engineers (NSBE)
- **Languages** – Fluent in English and Wolof; basic proficiency in French